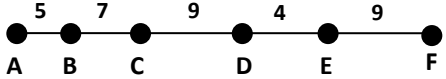
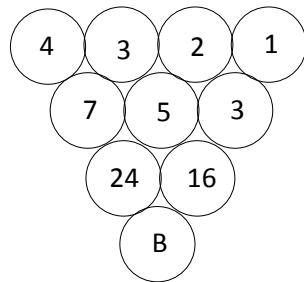


Q1	How many natural numbers between 1 and 100 contain <u>at least</u> one 5 but do not contain a 7?	Q2	Using the <u>same</u> rule to calculate each cell, find the value of C.
	5		5
Q3	A maths operation # is defined by $a \# b = a \times b - a$ so $3 \# 2 = 3 \times 2 - 3 = 3$ Find the value of $(4 \# 3) \# 2$	Q4	When one number is removed from the set 11 12 17 18 23 29 30 the average decreases by 1,5. Which number is removed?
	5		5
Q5	Find the <u>average</u> of the four prime numbers that are <u>closest</u> to 30.		
			5

Q6	In cooking 4 tablespoons = $\frac{1}{4}$ cup 1 cup = 250 ml. How many tablespoons are there in 2 LITRES?	Q7	A bag contains two blue socks and two black socks. If you select two socks at random, what is the probability (as a fraction in simplest form) that the socks are a matching pair?
	5		5
Q8	What is the <u>smallest</u> number that can be divided by each of the numbers 1,2,3,...,10?	Q9	I have an average of 68% for my first FOUR tests in Mathematics. What mark must I get in my fifth test so that my overall average is 70%?
	5		5
	Q10	The population of America is about 316 500 000. The world population is about 7 106 000 000. $\frac{1}{13}$ of all the people in the world are on Facebook and 71,2% of Americans are on Facebook. What % of non-Americans are on Facebook (<u>to NEAREST %</u>)?	
		5	

Q11	$20 \div 13 = 1,538461538461....$ has a recurring string of decimal digits. What digit is in position 2015 to the right of the decimal comma?	Q12	A train is 100 metres long. It is travelling at 30 km/h. The train enters a tunnel that is 50 metres long. How long (in seconds) will it take for the ENTIRE train to pass through the tunnel?
	10		10
Q13	The diagram below shows the distance between each of the houses of 6 friends A to F All 6 friends meeting at which house involves the <u>MOST</u> travel? 	Q14	A Maths operation @ is defined in the following examples $3 @ 6 = 9$ $4 @ 6 = 6$ $5 @ 2 = 1$ $3 @ 7 = 3$ So $3 @ 8 = ?$
	10		10

Q15



Using the same rule to calculate each cell, find the value of B.

10

Q16	Two trains are 150 metres apart on the <u>same track</u>. They are travelling towards each other. Train A is moving at 18 km/h. Train B is moving at 9 km/h. How many <u>seconds</u> will it be before they collide?	Q17	A 4-digit number has digits 1, 2, 3, 4. each used once. If <u>ALL</u> the 4-digit numbers that contain these digits are added together, what is the total?
	20		20
Q18	Find the SUM of the digits of $10^{20} - 63$	Q19	The odd numbers are arranged in a triangle $ \begin{array}{ccccccc} & & & 1 & & & \\ & & 3 & & 5 & & \\ & 7 & & 9 & & 11 & \\ 13 & & 15 & & 17 & & 19 \\ & & & \text{etc} & & & \end{array} $ What is the sum of the numbers in row 15?
	20		20
Q20	A number ladder is as follows $ \begin{array}{ccc} 1 & 1 & \\ 2 & 3 & \\ 5 & 7 & \\ 12 & 17 & \text{etc.} \end{array} $ The same rules are used to generate each row. Find the <u>sum</u> of the numbers in row 6.		
	20		

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ANSWERS

Q1 17	Q2 26	Q3 8	Q4 29	Q5 30	5 each
Q6 128	Q7 1/3	Q8 2520	Q9 78	Q10 5% (nb no decimal)	5 each
Q11 6	Q12 18 sec	Q13 F	Q14 6	Q15 320	10 each
Q16 20	Q17 66660	Q18 172	Q19 3375	Q20 169	20 each

Tour de Maths 2015
First leg – St Andrew's School for Girls



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First Name	Surname	School

SCORE

/200

Q1	Q2	Q3	Q4	Q5	5 marks each
Q6	Q7 (Simplest fraction)	Q8	Q9	Q10 (NB - nearest %)	5 marks each
Q11	Q12	Q13	Q14	Q15	10 marks each
Q16	Q17	Q18	Q19	Q20	20 marks each

SOLUTIONS

- Q1 10 numbers end in 5 (including 55), 9 start with 5 (excluding 55); remove 75 and 57
- Q2 add the two cells above to get the value in the cell below; $A = 14$, $B = 12$, $C = 26$
- Q3 $4\#3 = 12 - 4 = 8$ $8\#2 = 16 - 8 = 8$
- Q4 average of 7 numbers is 20 with a total of 140; $6 \times 18,5 = 111$, so remove 29
- Q5 4 primes are 29 & 31 and 23 and 37; average = 30
- Q6 8 cups in 2L; and $32 \frac{1}{4}$ cups so $32 \times 4 = 128$
- Q7 can get blue&blue or black&black each with prob of $\frac{1}{6}$, so adding gives $\frac{1}{3}$
OR select a sock at random then the prob you get the matching sock is $\frac{1}{3}$
OR there are 6 arrangements of 4 socks and only two of these start with same colour
- Q8 the number needs to be divisible by 10, 9, 4, 7 and product of these is 2520
- Q9 $4 \times 68 = 272$ and $5 \times 70 = 350$ and difference = 78
OR need to increase each of previous test by 2%
- Q10 world facebook = 546 615 384,6 (a) USA facebook = 225 348 000 (b)
world popn minus USA = 6 789 500 000 (c) $(a-b)/c \times 100$
- Q11 6 digits recur, $\text{int}(2015/6) = 335$; $335 \times 6 = 2010$ so 5th digit is 6
- Q12 end of train needs to travel 150m or 0,15 km time= $d/s = 0,15/30 \times 3600$
- Q13 total travelling for each venue is 97, 77, 63, 63, 71, 102; F worst
- Q14 multiply the two numbers together and then add the resulting digits
- Q15 subtract the squares of the numbers in the cells above
- Q16 train A is moving at 5 m/s and train B at 2,5 m/s; approaching at 7,5 m/s $150/7,5 = 20$
- Q17 24 numbers; 6 end in 1,2,3,4 total of each column is 60
- Q18 1 followed by 20 0's – 63 ends in 37 with 18 9's before
- Q19 sum of row n is n^3
- Q20 left hand number in each row obtained by adding two numbers in row above
right hand number obtained by adding left number + left number above that